Detecting AI Edits in Human Texts

Ege Çakar, Henry Rui Huang, Duy Thuc Nguyen

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Abstract

Most AI-assisted text is human-drafted and then partly edited by a model, so a document carries a sparse set of AI-edited sentences among human ones. We treat each edited sentence as a sparse signal and test documents with Higher Criticism (HC) on per-sentence conformal p-values. We build a corpus of 313k pre-2020 human sentences from eight domains, have six current LLMs (one held out) edit every train, val and test sentence under two prompts, and fine-tune sentence detectors on the edits. Our four-model ensemble detects 38.6% of edited test sentences at a 1% false-positive rate (36.2% for the held-out editor), against 8.7% for EditLens, the best open-source detector we evaluated, and 1.5–2.1% for zero-shot detectors. On whole documents of at least 16 sentences, HC with thresholds corrected for within-document correlation holds its level (type I error 0.045 at $\alpha = 0.05$) and rejects 92% of documents when 65% of sentences are edited ($\theta = 0.1$) and 22% when a single sentence is edited ($\theta = 0.9$). Work in progress covers a second test against fully AI-generated documents, pretraining language models on AI-edited text, third-party benchmarks, and further held-out editors.

1 Introduction

Text produced with AI today mostly comes from back-and-forth with a human and from polishing human drafts: student essays, peer reviews, emails, scientific writing. The result is mostly human, with an unknown number of AI-edited sentences scattered through it. Detectors for fully AI-generated text are well developed and widely used, but edited text is harder: human-written, AI-edited text follows a different distribution from fully generated text [1], and the edits that matter are often a few sentences in a long document.

This is a sparse-inference problem. A document of n sentences gives n tests, one per sentence, of $H_{0,i}$: sentence i is human-written. We place $m = \text{round}(n^{1-\theta})$ edited sentences in a document, so θ sets the sparsity; how strongly a sentence was edited (the share of its words changed, r) and how well a detector separates edited from human sentences set the signal strength. We test the global null H_0: every sentence of the document is human, with Higher Criticism [2, 3] and related global tests, then estimate which sentences and what fraction were edited.

Contributions:

- A paired corpus: 313,249 pre-2020 human sentences from eight sources, each train, val and test sentence edited by six LLMs under two prompts (about 2.0M accepted edits), split by document.
- Sentence detectors fine-tuned on the edits (29-run search, then full training of four backbones) and an ensemble, compared with four released detectors and six zero-shot ones.
- A document-level pipeline with exact per-sentence conformal p-values, global tests with thresholds that account for within-document correlation, support recovery and edited-fraction estimation, evaluated over the (θ, r) plane against the Gaussian rare/weak theory [4, 2].

2 Setup

Figure 1 shows the pipeline in three parts: building the data (a), turning one sentence into a p-value (b), and testing one document (c). Every choice below is recorded with its rationale in the repository's `docs/review.md`.

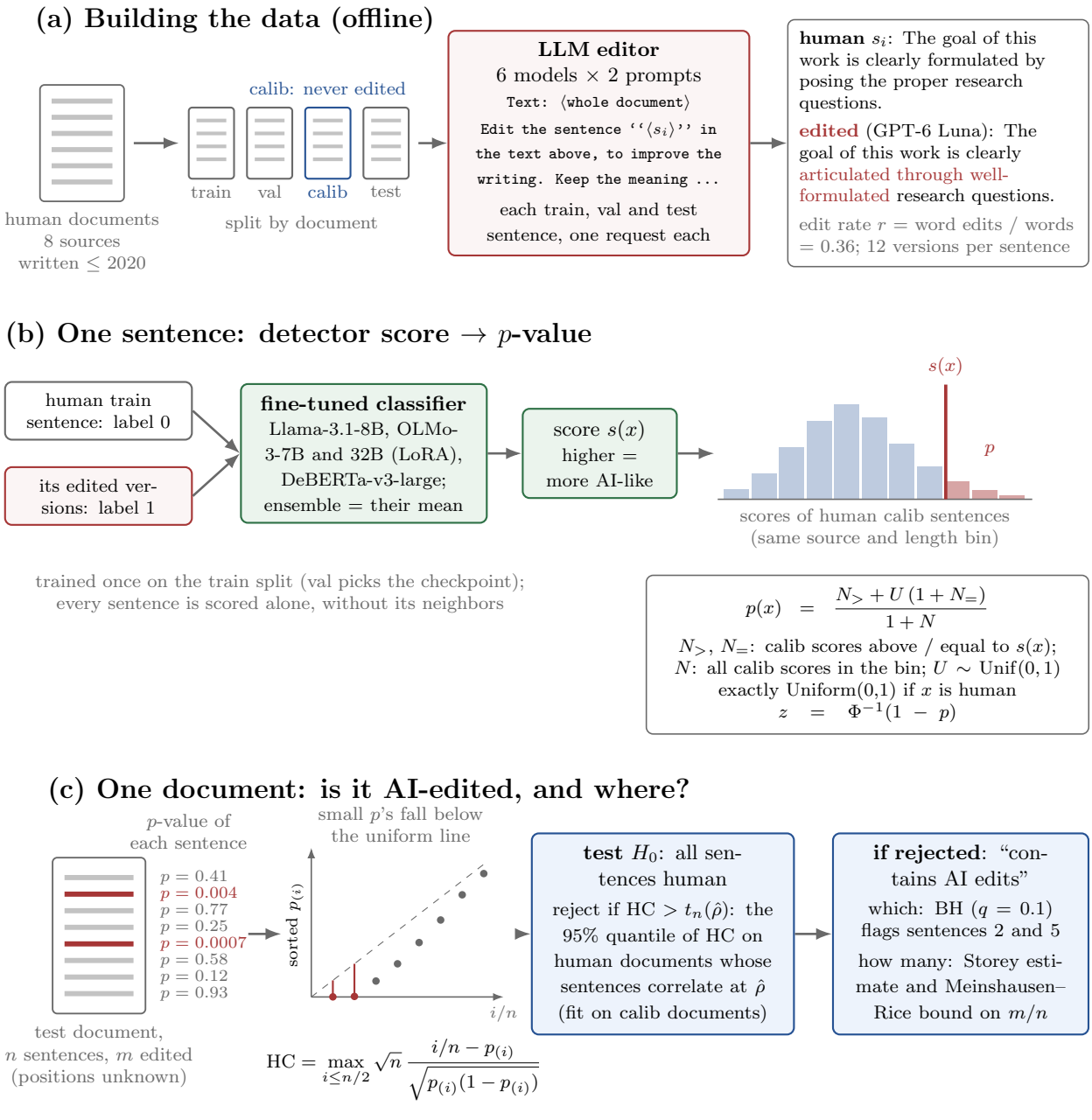


Figure 1. Pipeline. (a) Pre-2020 human documents are split by document; LLMs edit every train, val and test sentence one at a time while seeing the whole document; calib stays human. (b) A classifier trained on human vs. edited train sentences scores each sentence alone, and the score becomes an exact p-value by ranking it among human calib sentences. (c) A test document's n sentence p-values are combined by Higher Criticism (also min-p, HC$^+$ and Stouffer), with a threshold that allows for correlation between sentences of one document; BH then locates the edits. Evaluation: the 1,301 test documents with $n \geq 16$, each edited by every editor and prompt at $m = \mathrm{round}(n^{1-\theta})$ random positions for $\theta \in \{0.1, \ldots, 0.9\}$, and left unedited for the type I error.

	Documents				Median		Test docs
Source	train	val	calib	test	sentences/doc	Sentences	$n \geq 16$
FineWeb-Edu	386	31	364	218	27	39,412	149
peS2o	74	11	79	38	169	39,460	38
PERSUADE 2.0	713	74	690	349	20	39,034	241
ELLIPSE	784	81	787	376	18	39,027	231
XSum	828	69	779	404	15	39,046	203
OpenReview	681	73	652	344	19	39,081	207
Enron	1,885	181	1,880	1,010	5	39,067	77
WritingPrompts	305	32	301	175	40	39,122	155
Total	5,656	552	5,532	2,914	14	313,249	1,301

Table 1. Corpus. Test documents with at least 16 sentences form the document-level test set.

2.1 Human corpus

Eight sources cover web pages, papers, school essays, learner essays, news, peer reviews, emails and stories (Table 1): FineWeb-Edu [5], peS2o [6], PERSUADE 2.0 [7], ELLIPSE [8], XSum [9], OpenReview [10], Enron emails via AESLC [11, 12] and WritingPrompts [13]. Documents dated after 2020 are dropped where dates exist; the others predate it by construction. Each source gets about 15,000 train, 1,500 val, 15,000 calib and 7,500 test sentences, filled with whole documents, so splits are disjoint by document. Sentences come from Qwen3.5-122B [14] served with vLLM [15], accepted only when its output reproduces the input characters, with SaT [16] as the fallback.

2.2 AI edits

Each editable train, val and test sentence is edited once by each of six models: gpt-oss-120b [17], Gemma 4 31B [18], GPT-6 Luna [19], Claude Haiku 5.5 [20], GLM-5.3 Flash [21] (training editors) and Qwen3.5-122B-A10B [14] (held out: never seen by any detector we train or select). Calib stays human. The editor sees the whole original document and is asked to edit one sentence, under two prompts: "edit" (*Edit the sentence "…" in the text above to improve the writing. Keep the meaning and about the same length, and keep it one sentence.*) and "light" (*Lightly edit …, changing only what needs it*). Outputs that are empty, unchanged, split or merge sentences, alter an anonymization placeholder or more than double in length are resampled up to three times, then rejected. The edit rate r of a sentence is the word-level Levenshtein distance between the original and the edit divided by the original's word count. Table 2 lists the editors; median r ranges from 0.19 (GLM, light) to 0.50 (gpt-oss).

2.3 Sentence detectors

Fine-tuned detectors. Training uses every accepted train edit of the five training editors (about 1.08M) and the 120k human train sentences, with a class-weighted binary cross-entropy. Each sentence is scored in isolation, so a human sentence's score does not depend on whether its neighbors were edited and its p-value stays exactly uniform in any document. We searched 29 configurations on 20% of the training documents (Table 12): frozen embedders with a logistic-regression head (Qwen3-Embedding, EmbeddingGemma, Llama-Embed-Nemotron), fully fine-tuned encoders (DeBERTa-v3-large [22], ModernBERT-large, Ettin-encoder-1B [23]) and decoders with LoRA [24] and a classification head on the last token (Llama-3.1-8B [25], OLMo-3-7B [26], Ettin-decoder-1B). Selection metric: TPR at 1% FPR on the training editors' val sentences (12,335 human, 110,998 edited), the quantity that drives HC and Bonferroni. Fine-tuned models beat frozen embedders by a wide margin (0.30–0.32 vs. at most 0.17), and in every fine-tuned family the lowest learning rate won.

The three best families were then trained on all training documents for three epochs, at learning rates a factor of two around each search winner, keeping the best of ten val evaluations per epoch (Table 3). OLMo-3-32B [26] was added for one epoch, the time available. Decoders use LoRA rank 16, $\alpha = 32$, dropout

Editor (prompt)	Role	Accepted edits			Rejected	Edit rate r	
		train	val	test	(%)	median	mean
gpt-oss-120b	train	112,462	11,552	56,666	6.2	0.50	0.53
gpt-oss-120b (light)	train	111,618	11,496	56,257	6.8	0.36	0.40
gemma-4-31b	train	113,241	11,615	57,134	5.5	0.47	0.51
gemma-4-31b (light)	train	107,895	11,085	54,434	9.9	0.23	0.29
gpt-6-luna	train	110,629	11,343	55,808	7.7	0.38	0.43
gpt-6-luna (light)	train	106,058	10,941	53,542	11.4	0.23	0.28
claude-haiku-5.5	train	105,266	10,717	53,094	12.2	0.45	0.51
claude-haiku-5.5 (light)	train	92,471	9,450	46,447	22.9	0.20	0.26
glm-5.3-flash	train	113,382	11,622	57,263	5.3	0.33	0.40
glm-5.3-flash (light)	train	108,691	11,177	54,848	9.2	0.19	0.25
qwen3.5-122b	held out	110,397	11,257	55,637	7.9	0.39	0.43
qwen3.5-122b (light)	held out	105,619	10,813	53,157	11.9	0.25	0.29

Table 2. Editors and prompts. Rejected: share of requested edits still rejected after three attempts. Edit rate on the test split.

Backbone	Learning rates tried	Chosen	AUROC	TPR@0.1%	TPR@1%	TPR@5%
Llama-3.1-8B (LoRA)	2.5e-5 / 5e-5 / 1e-4: 0.351 / 0.350 / 0.326	2.5e-5	0.917	0.175	0.351	0.572
OLMo-3-7B (LoRA)	2.5e-5 / 5e-5 / 1e-4: 0.349 / 0.349 / 0.346	2.5e-5	0.912	0.181	0.349	0.567
DeBERTa-v3-large	2.5e-6 / 5e-6 / 1e-5: 0.337 / 0.342 / 0.334	5e-6	0.916	0.169	0.342	0.570
OLMo-3-32B (LoRA, 1 epoch)	5e-5	5e-5	0.923	0.178	0.374	0.609

Table 3. Full training runs: val TPR at 1% FPR for each learning rate tried, and val metrics of the chosen run (training editors' val sentences; the held-out editor is never used for selection). The best OLMo-3-7B checkpoint came in epoch 2. Full training adds about 3 points over the search runs (0.30–0.32).

0.05 on all linear layers; inputs are at most 256 tokens. The **ensemble** averages the four detectors' scores after standardizing each by its mean and standard deviation on val human sentences, so nothing is fit on calib or test.

Released detectors. EditLens [27] (Llama-3.2-3B with LoRA and a 4-bucket head; score = expected bucket), set up as in Pangram's release: 4-bit NF4 base [28], bf16 compute, Pangram's text cleaning, 1024 tokens. On 600 rows of Pangram's own test set our scores match their published ones with Pearson correlation 0.99999, mean absolute difference 0.0008 and the same predicted bucket on 597. desklib v1.01 [29] (DeBERTa-v3-large), GPTTrace v8s [30] (Ettin-1B) and GPTTrace Ettin-150M [31] (int8 ONNX, with its homoglyph normalization) are used as released. Each is applied two ways: to every sentence alone, which feeds the same p-value pipeline as our detectors, and to the whole document (truncated to the model's limit), with a document-level conformal test against calib human documents of the same source.

Zero-shot detectors. Fast-DetectGPT [32] (analytic), Binoculars [33] and mean token log-likelihood, each with two observer/performer pairs: Falcon-7B / Falcon-7B-Instruct [34] and Llama-3-8B / Llama-3-8B-Instruct [25].

2.4 From scores to p-values

For a test sentence of source k, let $\mathcal{N}_k$ be the calib human sentences of source k in the same word-count bin (10 quantile bins). With scores oriented so that higher means more AI-like, the randomized conformal

p-value is

$$p = \frac{\#\{s' \in \mathcal{N}_k : s' > s\} + U\,(1 + \#\{s' \in \mathcal{N}_k : s' = s\})}{1 + |\mathcal{N}_k|}, \qquad U \sim \mathrm{Unif}(0,1),$$

which is exactly $\mathrm{Unif}(0,1)$ for a human sentence exchangeable with $\mathcal{N}_k$ [35, 36], and $z = \Phi^{-1}(1-p)$. TPR at FPR α below means the share of edited test sentences with $p < \alpha$; the realized FPR on human test sentences is reported next to it.

2.5 Documents, global tests and estimation

Test documents with at least 16 sentences (1,301) are used whole. For each editor and prompt (12) and $\theta \in \{0.1, 0.2, \ldots, 0.9\}$, $m = \mathrm{round}(n^{1-\theta})$ positions are drawn uniformly and those sentences are replaced by their edits (a position whose edit was rejected stays human). Every draw is seeded by its cell, so all detectors see the same documents and positions (common random numbers). The 1,301 unedited documents measure type I error.

Global tests at level $\alpha = 0.05$: HC with $\alpha_0 = 1/2$,

$$\mathrm{HC}_n = \max_{1 \le i \le n/2} \sqrt{n}\, \frac{i/n - p_{(i)}}{\sqrt{p_{(i)}(1 - p_{(i)})}},$$

HC^+ (only $p_{(i)} \ge 1/n$) [2], min-p [37] and Stouffer's $\sum_i z_i/\sqrt{n}$ [38]. Sentences of one document share an author and topic, so their z's are positively correlated even when all are human. We model this as equicorrelation, $z_j = \sqrt{\rho}\, a + \sqrt{1-\rho}\, e_j$, estimate ρ per source and detector by the one-way ANOVA intraclass correlation of z on calib documents (cross-fitted: calib documents are split into two folds and each fold's p-values are computed against the other), and take each test's critical value at every n from 100,000 Monte Carlo draws of this null. $\hat{\rho}$ is 0.06–0.32 for our detectors. With i.i.d. thresholds HC's type I error is 0.083–0.104 instead of 0.05 (Table 7).

Which sentences were edited is estimated by Benjamini–Hochberg at $q = 0.1$ [39] and by Bonferroni at α, scored by Hamming error/m and exact recovery. The edited fraction $\varepsilon = m/n$ is estimated by Storey's estimator at $\lambda = 1/2$ [40] and bounded below by Meinshausen–Rice with the HC bounding function [41].

3 Results

3.1 Sentence level

Table 4 is the main comparison. Our ensemble detects 38.6% of edited sentences from the training editors and 36.2% from the held-out Qwen at 1% FPR; each of our single models detects 30–38%. The best released detectors reach 8.7% (EditLens) and 9.4% (GPTTrace v8s) on training editors, and the zero-shot detectors 1.5–2.1%, barely above the 1% they would reach by chance. Conformal calibration holds for every detector: the realized FPR on human test sentences is within 0.0005 of 0.1%, 0.001 of 1% and 0.004 of 5%.

Detection depends on the editor (Table 5) and, within an editor, on how much it changed (Table 6): the ensemble catches 66% of gpt-oss-120b's edits but 19% of GLM's light edits, and 19–23% of sentences with $r \le 0.1$ against 53% with $0.5 < r \le 1$. The held-out Qwen falls in the middle of the training editors, so there is no sign of overfitting to the training editors. GPTTrace v8s is the one released detector with a clear editor preference: it catches 25% of gpt-oss-120b edits at 0.1% FPR and at most 1.7% of any other editor's. Detection also varies by domain (Appendix, Table 13): from 22% on FineWeb-Edu to 61% on ELLIPSE learner essays, where editors fix many errors.

3.2 Document level

All detectors hold their level on human documents with the correlated thresholds (Table 7); min-p of OLMo-3-32B is the largest at 0.071. Table 8 gives HC power by θ. The ensemble rejects 92% of edited documents at $\theta = 0.1$ (on average 22.5 edited sentences, 65% of the document), 59% at $\theta = 0.5$ and 22% at $\theta = 0.9$, where a 16-sentence document has a single edited sentence. Held-out Qwen documents are detected at nearly the same rates. The released detectors are far behind, whether their sentence scores are combined with HC or

	Training editors			Qwen (held out)			Human: FPR		
Detector	0.1%	1%	5%	0.1%	1%	5%	0.1%	1%	5%
Ensemble	0.199	0.386	0.612	0.147	0.362	0.607	0.0011	0.0105	0.0495
OLMo-3-32B	0.193	0.377	0.599	0.140	0.352	0.596	0.0012	0.0111	0.0499
Llama-3.1-8B	0.169	0.341	0.571	0.103	0.295	0.546	0.0008	0.0102	0.0504
OLMo-3-7B	0.162	0.331	0.562	0.108	0.302	0.558	0.0010	0.0102	0.0496
DeBERTa-v3-large	0.170	0.341	0.573	0.121	0.316	0.565	0.0009	0.0101	0.0490
EditLens (3B)	0.018	0.087	0.243	0.022	0.098	0.251	0.0013	0.0108	0.0513
GPTTrace v8s (1B)	0.057	0.094	0.180	0.011	0.047	0.141	0.0009	0.0100	0.0504
desklib v1.01	0.002	0.022	0.083	0.003	0.024	0.086	0.0008	0.0102	0.0493
GPTTrace Ettin-150M	0.003	0.019	0.076	0.004	0.020	0.079	0.0010	0.0097	0.0503
Fast-DetectGPT (Llama-3-8B)	0.002	0.018	0.083	0.002	0.015	0.073	0.0014	0.0106	0.0514
Fast-DetectGPT (Falcon-7B)	0.002	0.015	0.068	0.001	0.012	0.058	0.0011	0.0095	0.0478
Log-likelihood (Llama-3-8B)	0.001	0.017	0.090	0.001	0.014	0.077	0.0009	0.0099	0.0490
Log-likelihood (Falcon-7B)	0.001	0.017	0.088	0.001	0.014	0.075	0.0008	0.0091	0.0471
Binoculars (Llama-3-8B)	0.003	0.021	0.088	0.002	0.018	0.078	0.0011	0.0106	0.0519
Binoculars (Falcon-7B)	0.003	0.017	0.074	0.002	0.014	0.063	0.0010	0.0096	0.0481

Table 4. Sentence level, test split: TPR (share of edited sentences with p below the FPR level) for training editors (545,493 edited sentences) and the held-out Qwen (108,794), and the realized FPR on 60,483 human test sentences. Blocks: ours (ensemble and its four members), released detectors, zero-shot detectors.

Editor (prompt)	Ensemble	OLMo-3-32B	Llama-3.1-8B	OLMo-3-7B	DeBERTa-v3-large	EditLens (3B)	GPTTrace v8s (1B)
gpt-oss-120b	0.661	0.651	0.604	0.598	0.613	0.133	0.308
gpt-oss-120b (light)	0.576	0.568	0.537	0.524	0.535	0.095	0.286
gemma-4-31b	0.449	0.436	0.358	0.379	0.392	0.126	0.065
gemma-4-31b (light)	0.258	0.246	0.217	0.216	0.218	0.061	0.030
gpt-6-luna	0.438	0.431	0.405	0.363	0.362	0.096	0.057
gpt-6-luna (light)	0.313	0.307	0.296	0.257	0.259	0.067	0.042
claude-haiku-5.5	0.417	0.403	0.348	0.342	0.365	0.090	0.040
claude-haiku-5.5 (light)	0.242	0.233	0.211	0.202	0.213	0.061	0.025
glm-5.3-flash	0.283	0.273	0.234	0.235	0.250	0.075	0.035
glm-5.3-flash (light)	0.191	0.183	0.169	0.161	0.168	0.054	0.026
qwen3.5-122b	0.425	0.415	0.341	0.354	0.373	0.118	0.058
qwen3.5-122b (light)	0.295	0.286	0.248	0.247	0.255	0.076	0.036

Table 5. TPR at 1% FPR by editor and prompt, test split. Qwen rows are held out.

the whole document is scored directly: EditLens reaches 0.46–0.56 at $\theta = 0.1$ and at most 0.18 from $\theta = 0.5$ on. The zero-shot detectors reach at most 0.13 at $\theta = 0.1$. For EditLens, HC over sentence scores beats scoring the whole document from $\theta = 0.3$ upward, which is where the sparse regime starts.

Per editor (Table 9), HC power at $\theta = 0.1$ ranges from 0.76 (Claude Haiku, light) to 0.999 (gpt-oss-120b), and at $\theta = 0.9$ from 0.10 to 0.45. Min-p is better than HC in the sparsest cells ($\theta \geq 0.7$) and worse in dense ones; Stouffer is below HC everywhere, closest at $\theta = 0.1$ (0.87 vs. 0.92) (Appendix, Tables 14 and 15). Figure 2 places each document in the (θ, r) plane at its effective signal strength $r_{\text{eff}} = \max(\mu_{\text{eff}}, 0)^2/(2 \log n)$, where μ_{eff} is the median z of edited sentences: empirical power tracks the Gaussian simulation at the same n.

Recovery and edited fraction. Locating the edits is much harder than detecting them (Table 10). BH's Hamming error is $0.46m$ at $\theta = 0.1$ and $0.87m$ at $\theta = 0.9$; exact recovery happens in 4–28% of documents, the highest at $\theta = 0.9$, where there is one sentence to find. Storey's estimate is close to the true fraction for dense edits (0.62 vs. 0.65 at $\theta = 0.1$) and biased upward when edits are sparse (0.19 vs. 0.04 at $\theta = 0.9$). The Meinshausen–Rice bound covers the true fraction in 97–100% of documents, as its 95% guarantee requires, but is loose (0.23 at $\theta = 0.1$).

Edit rate r	Sentences	Ensemble	OLMo-3-32B	EditLens (3B)
≤ 0.05	12,391	0.225	0.214	0.035
0.05–0.1	53,355	0.190	0.183	0.035
0.1–0.2	121,871	0.248	0.242	0.046
0.2–0.3	110,141	0.337	0.329	0.067
0.3–0.5	196,830	0.443	0.432	0.097
0.5–1	148,440	0.533	0.521	0.153
> 1	11,259	0.319	0.305	0.075

Table 6. TPR at 1% FPR by edit rate r, all editors pooled. $r > 1$: edits longer than a full rewrite of the original's word count.

	i.i.d. thresholds		Equicorrelated thresholds				
Detector	HC	min-p	HC	HC$^+$	min-p	Stouffer	$\hat{\rho}$ (by source)
Ensemble	0.085	0.046	0.045	0.038	0.052	0.042	0.07–0.32
OLMo-3-32B	0.095	0.058	0.054	0.040	0.071	0.038	0.06–0.32
Llama-3.1-8B	0.083	0.043	0.048	0.048	0.049	0.042	0.06–0.30
OLMo-3-7B	0.088	0.047	0.048	0.048	0.054	0.044	0.07–0.31
DeBERTa-v3-large	0.088	0.048	0.041	0.038	0.053	0.041	0.06–0.29
EditLens (3B)	0.104	0.053	0.058	0.051	0.056	0.041	0.08–0.28
GPTTrace v8s (1B)	0.069	0.036	0.049	0.063	0.040	0.050	0.06–0.13
desklib v1.01	0.066	0.048	0.053	0.057	0.049	0.049	0.04–0.12
GPTTrace Ettin-150M	0.084	0.047	0.058	0.058	0.048	0.058	0.04–0.16
Fast-DetectGPT (Llama-3-8B)	0.076	0.058	0.054	0.048	0.063	0.045	0.05–0.13
Fast-DetectGPT (Falcon-7B)	0.066	0.054	0.051	0.047	0.055	0.043	0.04–0.11
Log-likelihood (Llama-3-8B)	0.075	0.039	0.046	0.045	0.044	0.037	0.08–0.29
Log-likelihood (Falcon-7B)	0.078	0.035	0.044	0.046	0.039	0.036	0.07–0.29
Binoculars (Llama-3-8B)	0.078	0.052	0.058	0.055	0.052	0.055	0.05–0.12
Binoculars (Falcon-7B)	0.065	0.041	0.050	0.054	0.041	0.046	0.04–0.11

Table 7. Type I error on the 1,301 human test documents at $\alpha = 0.05$, with i.i.d. and equicorrelated thresholds, and the range of $\hat{\rho}$ over sources.

4 Work in progress

4.1 Human-drafted vs. fully generated documents

Stage 1 rejects H_0 for fully AI-generated documents too, so we add a second test of H_0': every sentence is AI. GPT-6 Luna wrote a one-line topic for each of the 8,446 calib and test documents, and five generators wrote a document per topic of the same genre and length without seeing the human text. Stage-2 p-values rank each sentence against generated calib sentences of four generators (lower tail: "more human than generated text"), with correlated thresholds at $\hat{\rho}' = 0.05$–0.19; Qwen is held out. Results with the three-model ensemble (Llama, OLMo-3-7B, DeBERTa):

- Stage 1 rejects 98.7–100% of generated test documents.
- H_0' type I error, four calibration generators pooled (5,179 documents): HC 0.050, min-p 0.045, Stouffer 0.036. Per generator it ranges from 0.012 (Luna) to 0.082 (Gemma), so the level holds only for the mixture.
- Held-out Qwen: HC and min-p over-reject (0.125, 0.131); Stouffer holds (0.052).
- Power: H_0' is rejected for 99.5% of human documents and 95–99% of sparse-edit documents. Calling a document "human-drafted, AI-edited" needs both stages to reject, so it is limited by stage 1: 0.87 at $\theta = 0.1$, 0.21 at $\theta = 0.9$.

Detector	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	0.92/0.93	0.78/0.78	0.59/0.56	0.41/0.38	0.22/0.19
OLMo-3-32B	0.91/0.93	0.77/0.77	0.58/0.55	0.41/0.38	0.22/0.19
Llama-3.1-8B	0.89/0.89	0.72/0.69	0.53/0.47	0.37/0.30	0.20/0.16
OLMo-3-7B	0.89/0.91	0.72/0.72	0.52/0.49	0.36/0.31	0.19/0.16
DeBERTa-v3-large	0.90/0.92	0.75/0.76	0.55/0.53	0.38/0.35	0.21/0.18
EditLens (3B)	0.46/0.50	0.28/0.31	0.17/0.18	0.12/0.14	0.08/0.08
GPTTrace v8s (1B)	0.35/0.29	0.26/0.17	0.19/0.11	0.14/0.08	0.09/0.06
desklib v1.01	0.14/0.14	0.09/0.10	0.07/0.08	0.06/0.06	0.06/0.06
GPTTrace Ettin-150M	0.13/0.14	0.09/0.11	0.07/0.08	0.07/0.07	0.06/0.06
Fast-DetectGPT (Llama-3-8B)	0.11/0.09	0.08/0.07	0.07/0.06	0.06/0.06	0.06/0.06
Fast-DetectGPT (Falcon-7B)	0.09/0.07	0.07/0.06	0.06/0.06	0.06/0.05	0.05/0.05
Log-likelihood (Llama-3-8B)	0.11/0.08	0.07/0.06	0.05/0.05	0.05/0.05	0.05/0.05
Log-likelihood (Falcon-7B)	0.10/0.07	0.06/0.05	0.05/0.05	0.05/0.05	0.04/0.04
Binoculars (Llama-3-8B)	0.13/0.10	0.09/0.08	0.07/0.07	0.07/0.06	0.06/0.06
Binoculars (Falcon-7B)	0.10/0.07	0.07/0.06	0.06/0.06	0.06/0.05	0.05/0.05
EditLens (3B) (whole doc.)	0.53/0.56	0.25/0.23	0.13/0.11	0.08/0.07	0.06/0.05
GPTTrace v8s (1B) (whole doc.)	0.39/0.26	0.21/0.10	0.13/0.07	0.10/0.06	0.07/0.06
desklib v1.01 (whole doc.)	0.35/0.34	0.18/0.15	0.11/0.09	0.08/0.06	0.06/0.05
GPTTrace Ettin-150M (whole doc.)	0.09/0.08	0.06/0.06	0.05/0.05	0.05/0.05	0.05/0.05

Table 8. HC rejection rate on edited whole test documents ($n \geq 16$), equicorrelated thresholds; each cell is training editors / held-out Qwen. The last block scores each whole document with the released detector directly (document-level conformal test against calib human documents, type I error 0.050–0.055).

- A composite null, "at least 90% of sentences AI", rejected when the Meinshausen–Rice bound on the human fraction exceeds 0.1, has type I error at most 0.011 and power 0.985 on human documents.

Open: per-generator validity, and a second held-out generator (Claude Sonnet 5.5 was skipped).

4.2 Training language models on AI-edited text

Does AI-edited text help or hurt as training data, and can per-sentence weights recover the human distribution? We pretrain small Llama-style models from scratch with the Pythia tokenizer [42] on the corpus's train, val and calib documents (5.8M tokens), at a Chinchilla-optimal size [43] (0.3M non-embedding parameters) and a data-starved one (1.6M), three seeds each, and evaluate perplexity on the human test documents. Arms: human data; a "mixture" where each document gets edits from one random training editor at a random θ; "all-edited" copies where every sentence is replaced by one editor's edit (1 pass or all 10 editor-prompt passes); and controls with the same token counts (human repeated 10 times, and 26M tokens of fresh human documents from the same sources [44]). Oracle weights per sentence: *mask* (0 for edited sentences) and *strength* ($1 - \min(r, 1)$). The paired Δ test scores 30,000 test edits against their human originals: Δ is the mean over pairs of the per-token log-probability of the edit minus that of the original, so $\Delta > 0$ means the model prefers the AI-edited version [45].

Findings so far (Table 11): at matched tokens, AI-edited text is nearly as useful for perplexity as the human text it came from (10 passes: 156.7 vs. 155.5 for human repeated), and fresh human text is much better (118.0). Oracle weights never beat unweighted training on perplexity; masking discards most tokens and is far worse. The Δ test shows the cost of AI-edited data: training on all-edited text makes the model prefer AI-edited sentences ($\Delta = +0.38$ vs. -0.03 for human repeated), strength weighting removes about 10% of that shift, and masking removes all of it at the cost of perplexity. Because these models are data-starved, we are now repeating every arm as LoRA continued pretraining of released models (Pythia 410M–6.9B, OLMo-2-1B base and Instruct [46]; 28 jobs submitted). Before any training, Pythia-410M and 1B already prefer the edited sentence by $\Delta \approx +0.26$, so the quantity of interest there is the shift from that baseline.

Editor (prompt)	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
gpt-oss-120b	1.00	0.99	0.93	0.76	0.45
gpt-oss-120b (light)	1.00	0.97	0.89	0.68	0.39
gemma-4-31b	0.98	0.90	0.70	0.48	0.24
gemma-4-31b (light)	0.87	0.65	0.39	0.25	0.13
gpt-6-luna	0.98	0.89	0.71	0.51	0.27
gpt-6-luna (light)	0.93	0.75	0.53	0.34	0.18
claude-haiku-5.5	0.97	0.84	0.62	0.40	0.20
claude-haiku-5.5 (light)	0.76	0.51	0.33	0.22	0.12
glm-5.3-flash	0.92	0.74	0.47	0.29	0.16
glm-5.3-flash (light)	0.78	0.52	0.33	0.21	0.10
qwen3.5-122b	0.97	0.86	0.67	0.46	0.24
qwen3.5-122b (light)	0.90	0.70	0.46	0.29	0.14
All editors	0.92	0.78	0.59	0.41	0.22

Table 9. Ensemble HC rejection rate by editor and prompt, whole test documents.

θ	$\bar{m}$	ε	$\hat{\varepsilon}_{\text{Storey}}$	$\hat{\varepsilon}_{\text{MR}}$	MR covers	BH Hamming/m	BH exact	Bonf. Hamming/m	Bonf. exact
0.1	22.51	0.65	0.62	0.23	1.00	0.46	0.05	0.78	0.00
0.2	15.48	0.47	0.46	0.14	1.00	0.53	0.04	0.78	0.00
0.3	10.70	0.34	0.35	0.09	1.00	0.59	0.05	0.78	0.01
0.4	7.38	0.24	0.29	0.05	0.99	0.64	0.06	0.78	0.01
0.5	5.11	0.17	0.25	0.03	0.99	0.68	0.07	0.79	0.02
0.6	3.61	0.12	0.22	0.02	0.99	0.73	0.09	0.79	0.04
0.7	2.54	0.09	0.21	0.02	0.98	0.77	0.12	0.79	0.08
0.8	1.86	0.07	0.20	0.01	0.98	0.81	0.16	0.79	0.11
0.9	1.00	0.04	0.19	0.01	0.97	0.87	0.28	0.77	0.26

Table 10. Ensemble, whole test documents: mean number of edited sentences, true and estimated edited fraction, Meinshausen–Rice coverage, and BH ($q = 0.1$) and Bonferroni ($\alpha = 0.05$) support recovery.

4.3 Other ongoing work

- **Third-party benchmarks.** Running our detectors and pipeline on the public datasets that EditLens and desklib report on, and on other AI-edited and sentence-level benchmarks (e.g. RAID [47], M4 [48], MAGE [49]); data are being prepared.
- **A second held-out editor.** Claude Sonnet 5.5 [50] edited 20,000 sentences (6,265 test) under both prompts; scoring them tests generalization to an unseen editor from a training editor's family. We also plan to fine-tune the trained detectors on these edits.
- **Document-context detectors.** Detectors that score every sentence of a document in one pass (ModernBERT-large, Ettin-encoder-1B, Llama-3.1-8B reading the document twice) are configured but not trained. A sentence's score then depends on its neighbors, so the global null needs calib documents rather than calib sentences.
- **To do: PC-adjusted HC.** The equicorrelated thresholds fix HC's level, but a shared document effect still costs power. PC-adjusted HC removes the top K principal components of the within-document covariance of z (to be estimated on calib documents) before computing HC. In a simulation with one style factor ($\Sigma = \mathbf{1}\mathbf{1}^\top + I$, $n = 32$, $m = 6$) it has type I error 0.048 where plain HC with i.i.d. thresholds has 0.120, and power 0.47 vs. 0.16 at $r = 0.5$ (both at exact null thresholds). Open: estimating the covariance from calib documents of varying length, choosing K, and its power loss when edits are dense (the top component then absorbs signal). Innovated HC [51] is the analogous to-do for correlation between neighboring sentences.
- **Other released detectors.** ImBD [52] and DetectAnyLLM [53] are candidates; a literature search found no open detector reported to beat EditLens on edited text.

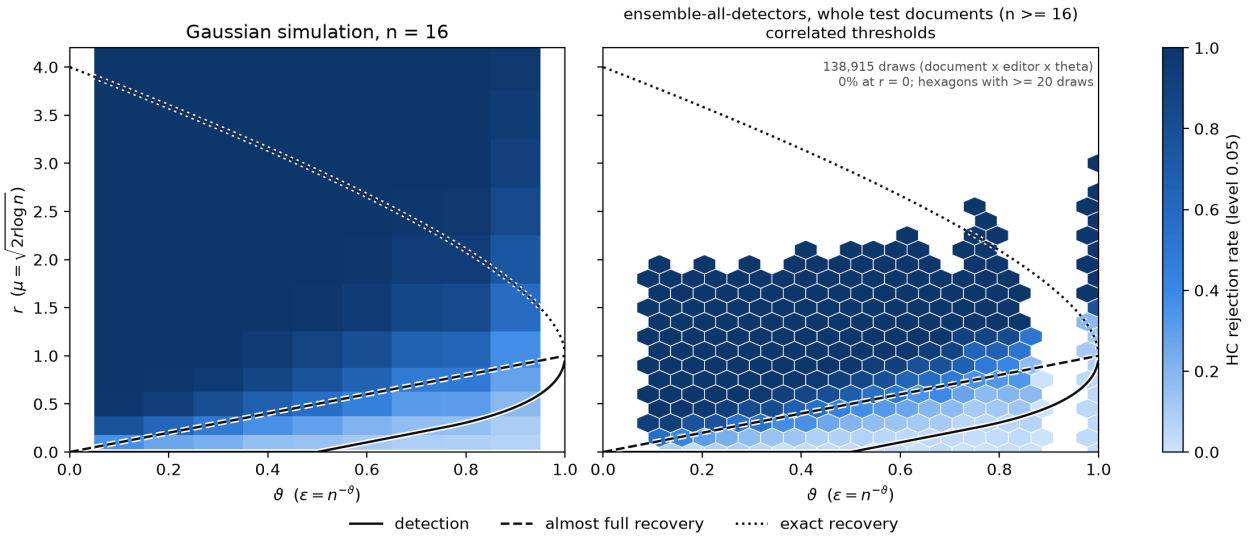


Figure 2. Left: HC power in the Gaussian model at $n = 16$ with the detection, almost-full recovery and exact recovery boundaries. Right: the ensemble on whole test documents, each (document, editor, θ) placed at its effective r.

5 Limitations

Edited documents are assembled from independent single-sentence edits, each made against the original document; a real AI pass edits sentences jointly. Edit positions are uniform, while real edits are not. The null is per source, so the domain is assumed known at test time. Equicorrelation is the simplest dependence model; HC^+ of GPTTrace v8s and min-p of OLMo-3-32B reach 0.063 and 0.071. Released detectors are applied to single sentences, which they were not trained for; their whole-document results are reported for that reason. The pretraining results come from very small models.

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Arm	Train tokens	Test perplexity		Δ (nats/token)	
		Chinchilla	data-starved	Chinchilla	data-starved
human	5.8M	341.0 ± 5.0	268.6 ± 5.0	−0.122	−0.104
mixture, unweighted	5.8M	341.1 ± 6.4	266.5 ± 2.2	−0.037	−0.015
mixture, strength	5.8M	342.8 ± 6.7	266.7 ± 1.0	−0.047	−0.030
mixture, mask	5.8M	353.1 ± 5.9	280.2 ± 2.5	−0.103	−0.091
human, 1 pass (train + val)	3.1M	556.6 ± 16.4	453.0 ± 19.5	−0.145	−0.141
all-edited 1 pass, unweighted	3.0M	600.7 ± 12.5	491.9 ± 4.3	+0.042	+0.070
all-edited 1 pass, strength	3.0M	603.1 ± 14.5	504.9 ± 9.5	+0.022	+0.044
all-edited 1 pass, mask	3.0M	1265.3 ± 20.0	1113.0 ± 9.0	−0.160	−0.159
human repeated 10×	30.8M	193.5 ± 9.7	155.5 ± 0.4	−0.056	−0.025
all-edited 10 passes, unweighted	29.7M	196.5 ± 1.6	156.7 ± 1.3	+0.312	+0.382
all-edited 10 passes, strength	29.7M	199.3 ± 1.3	159.2 ± 0.5	+0.285	+0.344
all-edited 10 passes, mask	29.7M	529.7 ± 6.8	469.5 ± 2.9	−0.139	−0.124
human fresh	26.2M	161.7 ± 1.2	118.0 ± 1.3	−0.026	+0.002

Table 11. Pretraining from scratch: test perplexity on human test documents (mean ± sd over 3 seeds) and the paired Δ test (sd over seeds 0.001–0.011). Blocks: matched tokens to the human corpus; one pass over train and val documents; about 30M tokens.

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A Additional results

Run (model, lr, class weighting)	AUROC	TPR@0.1%	TPR@1%	TPR@5%
llama-3.1-8b_lr5e-05	0.908	0.126	0.321	0.547
olmo-3-7b_lr5e-05	0.899	0.149	0.320	0.520
deberta-v3-large_lr5e-06_per_sentence	0.902	0.130	0.301	0.522
deberta-v3-large_lr5e-06_global	0.905	0.128	0.299	0.520
deberta-v3-large_lr1e-05_per_sentence	0.902	0.128	0.298	0.522
deberta-v3-large_lr2e-05_global	0.894	0.112	0.287	0.493
deberta-v3-large_lr2e-05_per_sentence	0.894	0.105	0.287	0.493
olmo-3-7b_lr0.0002	0.891	0.140	0.286	0.492
deberta-v3-large_lr1e-05_global	0.902	0.129	0.282	0.509
ettin-encoder-1b_lr1e-05_global	0.900	0.135	0.279	0.509
ettin-encoder-1b_lr1e-05_per_sentence	0.898	0.139	0.278	0.500
ettin-encoder-1b_lr2e-05_global	0.894	0.130	0.277	0.492
ettin-decoder-1b_lr5e-05	0.889	0.127	0.274	0.489
ettin-encoder-1b_lr2e-05_per_sentence	0.892	0.127	0.271	0.490
modernbert-large_lr3e-05_global	0.887	0.120	0.268	0.469
modernbert-large_lr1e-05_global	0.892	0.106	0.264	0.481
ettin-decoder-1b_lr0.0002	0.889	0.133	0.263	0.491
modernbert-large_lr1e-05_per_sentence	0.890	0.112	0.261	0.478
modernbert-large_lr3e-05_per_sentence	0.887	0.116	0.259	0.471
modernbert-large_lr5e-05_global	0.883	0.115	0.249	0.466
modernbert-large_lr5e-05_per_sentence	0.877	0.111	0.247	0.451
ettin-encoder-1b_lr4e-05_global	0.878	0.110	0.244	0.451
ettin-encoder-1b_lr4e-05_per_sentence	0.877	0.114	0.238	0.451
llama-3.1-8b_lr0.0002	0.851	0.087	0.187	0.396
llama-embed-nemotron-8b	0.832	0.075	0.174	0.363
qwen3-emb-8b	0.827	0.041	0.140	0.347
qwen3-emb-0.6b	0.784	0.056	0.121	0.277
qwen3-emb-4b	0.793	0.017	0.094	0.265
embeddinggemma-2	0.710	0.012	0.059	0.172

Table 12. Detector search on 20% of training documents, 2 epochs, val metrics on the training editors. Run names: model, learning rate, class weighting (global: inverse class frequency; per_sentence: each human sentence and the set of its edits weigh the same).

	Ensemble		OLMo-3-32B		EditLens (3B)	
Source	train	Qwen	train	Qwen	train	Qwen
FineWeb-Edu	0.222	0.193	0.218	0.193	0.052	0.067
peS2o	0.434	0.375	0.432	0.382	0.046	0.058
PERSUADE 2.0	0.442	0.428	0.438	0.419	0.109	0.121
ELLIPSE	0.611	0.600	0.618	0.607	0.231	0.229
XSum	0.368	0.351	0.353	0.333	0.051	0.068
OpenReview	0.415	0.427	0.398	0.410	0.081	0.116
Enron	0.334	0.288	0.303	0.257	0.054	0.055
WritingPrompts	0.243	0.200	0.228	0.183	0.058	0.056

Table 13. TPR at 1% FPR by source, test split.

Detector	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	0.83/0.84	0.73/0.72	0.59/0.56	0.45/0.42	0.27/0.23
OLMo-3-32B	0.83/0.84	0.72/0.72	0.59/0.56	0.46/0.42	0.28/0.25
Llama-3.1-8B	0.78/0.75	0.67/0.62	0.54/0.48	0.41/0.34	0.24/0.19
OLMo-3-7B	0.79/0.79	0.68/0.64	0.53/0.49	0.39/0.35	0.23/0.19
DeBERTa-v3-large	0.81/0.82	0.70/0.70	0.56/0.54	0.42/0.39	0.25/0.21
EditLens (3B)	0.35/0.41	0.25/0.29	0.17/0.20	0.13/0.15	0.09/0.09

Table 14. Min-p rejection rate, whole test documents, training editors / Qwen.

Detector	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	0.87/0.88	0.45/0.43	0.18/0.17	0.10/0.09	0.06/0.06
OLMo-3-32B	0.87/0.89	0.45/0.44	0.18/0.18	0.09/0.09	0.06/0.05
Llama-3.1-8B	0.86/0.85	0.43/0.40	0.17/0.16	0.09/0.08	0.06/0.05
OLMo-3-7B	0.85/0.86	0.41/0.40	0.17/0.17	0.09/0.09	0.06/0.06
DeBERTa-v3-large	0.87/0.88	0.44/0.44	0.18/0.18	0.10/0.09	0.06/0.06
EditLens (3B)	0.38/0.40	0.16/0.16	0.08/0.09	0.06/0.06	0.05/0.05

Table 15. Stouffer rejection rate, whole test documents, training editors / Qwen.

Detector	n	Windows	Type I	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	16	1,301	0.053	0.89	0.77	0.61	0.42	0.27
	32	443	0.052	0.90	0.72	0.54	0.37	0.18
OLMo-3-32B	16	1,301	0.052	0.88	0.76	0.60	0.42	0.26
	32	443	0.056	0.89	0.71	0.54	0.36	0.18
Llama-3.1-8B	16	1,301	0.045	0.85	0.72	0.55	0.37	0.24
	32	443	0.056	0.86	0.65	0.47	0.32	0.16
OLMo-3-7B	16	1,301	0.054	0.85	0.72	0.55	0.36	0.23
	32	443	0.045	0.87	0.67	0.49	0.31	0.16
DeBERTa-v3-large	16	1,301	0.049	0.87	0.74	0.58	0.39	0.25
	32	443	0.045	0.88	0.68	0.50	0.34	0.17

Table 16. Fixed-length windows: one random window of n consecutive sentences per test document with at least n; HC type I error and power, equicorrelated thresholds.